DO NOW — RECALL

Lesson 092 · The Three Buses: City Highways

Name: _____ Class: _____ Date: _____

1. What did PC advance to after sequential instruction at20?

2. What is MDR used for?

3. Which unit emits control signals?

4. Convert binary 10110000_2 to hexadecimal.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

21.

2. Model answer

Data/instruction transfer with memory.

3. Model answer

Control Unit.

4. Model answer

$1011\ 0000_2 = B0_{16}$.

By the end, you can…

• Address bus identifies the location the processor wants to access; typically the CPU drives address lines on a simple memory read.

• Data bus transfers the contents between memory and processor; for ordinary memory reads the bits travel memory-to-CPU.

• Control bus signals operation and coordination, including read/write; distinguish purpose and direction rather than treating all buses as one road.

Vocabulary — know these terms

address bus — Identifies location.

data bus — Transfers content bits.

control bus — Signals requests/status.

memory read — Retrieve contents of addressed location.

NEW LEARNING

Read, annotate and explain

Observe

Address bus identifies the location the processor wants to access; typically the CPU drives address lines on a simple memory read.

Explain

Data bus transfers the contents between memory and processor; for ordinary memory reads the bits travel memory-to-CPU.

Evaluate

Control bus signals operation and coordination, including read/write; distinguish purpose and direction rather than treating all buses as one road.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

CPU reads memory[12]=99. Label each bus with what it carries.

Explain the mechanism

Address bus12; control bus READ; data bus returns99, placed in MDR.

Address12 and content99 are different types of information.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

A read from address7 returns value150. Identify address/data/control signals. [6]

Question 2

Explain role and typical direction of all three buses for a write to memory address20 of value9. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate why confusing addresses with contents can break CPU fetch models; use PC,MAR,MDR,CIR in example. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

The address bus and data bus always carry the same binary number.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.